

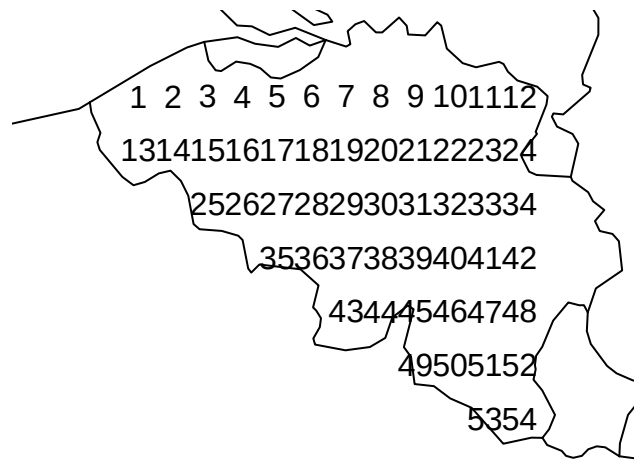
Multivariate extremes

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UCLouvain and KU Leuven, October 2024

Temperature data in Belgium

Plotting a map of Belgium with the grid points for which we have temperature data:



Closest stations:

- Louvain-la-Neuve: around 30
- Leuven: around 20

Some elementary univariate analysis

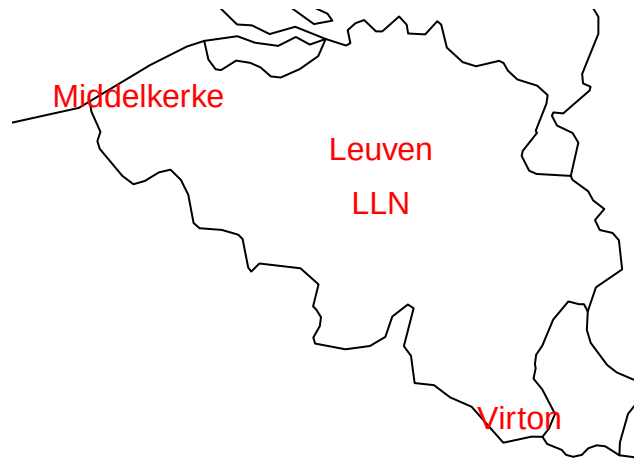
The temperature data are annual maxima, so it makes sense to fit a GEV distribution to them. We do so for every grid point. The number of observations (rows) and number of grid points (columns):

```
## [1] 69 54
```

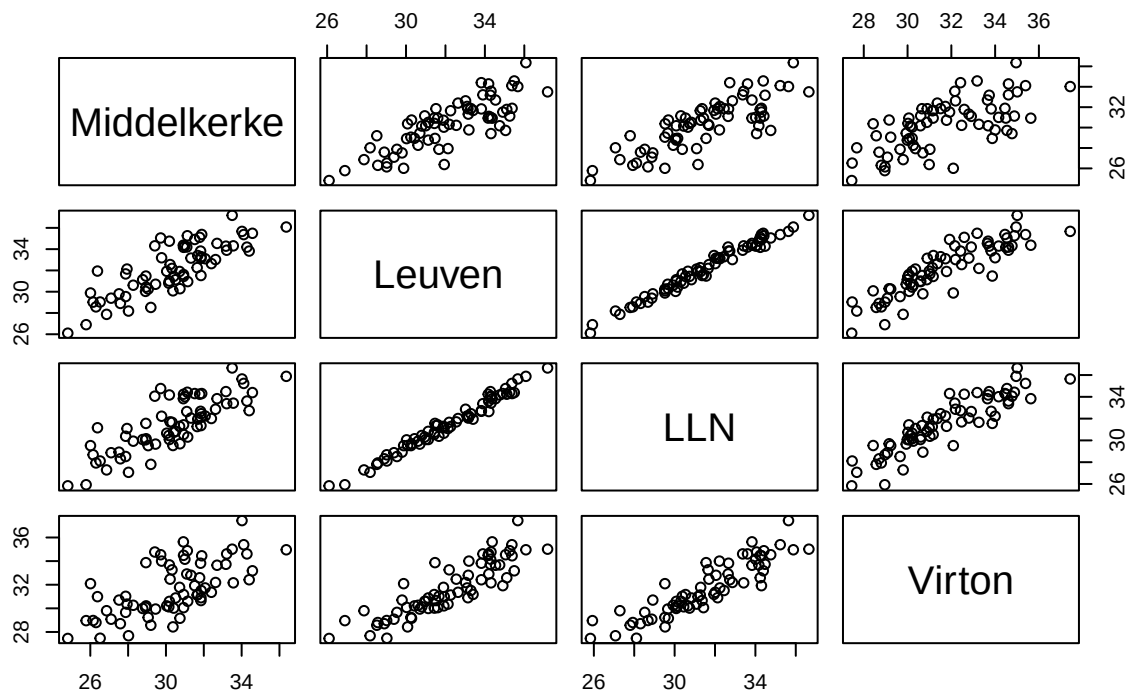
Fitting the GEV distribution to the annual temperature maxima in each grid point:

##	loc	scale	shape
##	Min. :28.83	Min. :2.072	Min. :-0.49747
##	1st Qu.:29.89	1st Qu.:2.246	1st Qu.: -0.36874
##	Median :30.42	Median :2.413	Median :-0.30962
##	Mean :30.54	Mean :2.399	Mean :-0.30582
##	3rd Qu.:31.12	3rd Qu.:2.500	3rd Qu.: -0.25835
##	Max. :32.46	Max. :2.737	Max. :-0.07292

Four cities in Belgium



Maximum temperatures in Belgium, 1950–2018

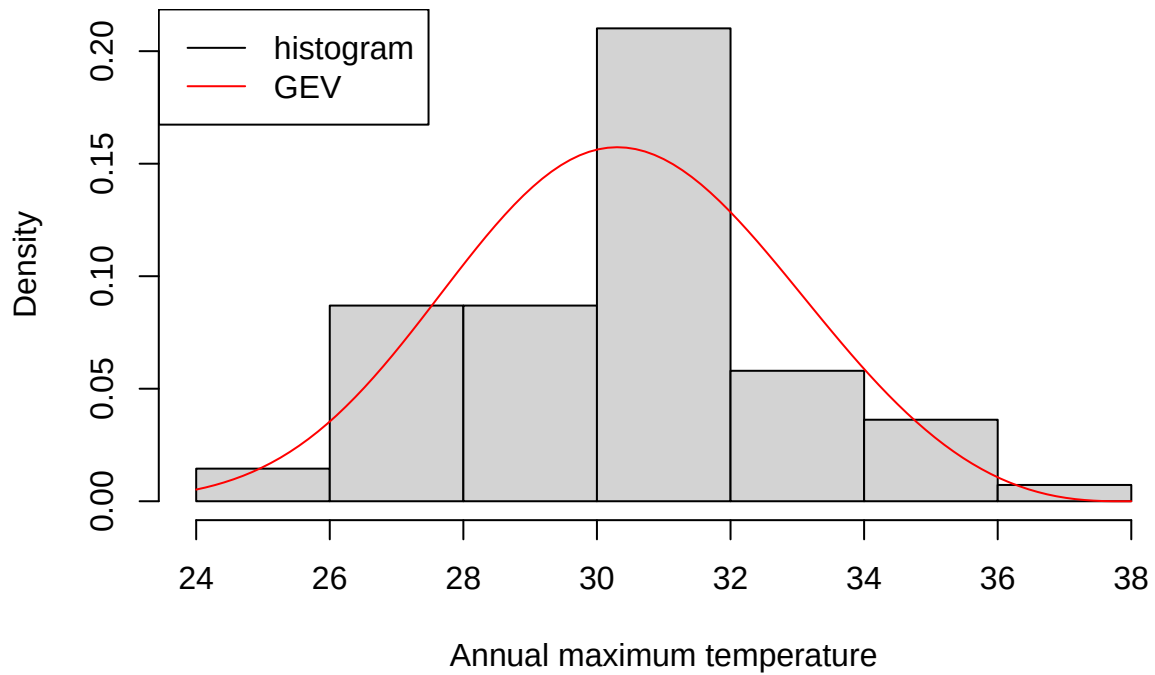


Fitting univariate GEV distributions to the annual maxima data in each city:

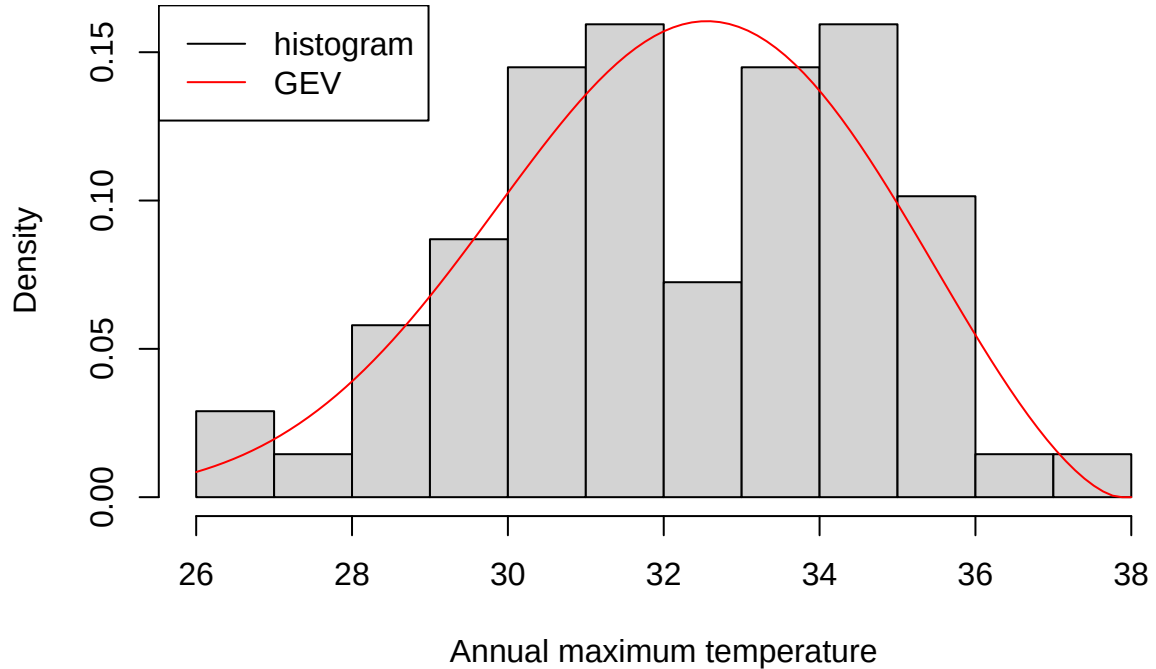
```
##          loc    scale    shape
## Middelkerke 29.49179 2.454463 -0.2946827
## Leuven      31.42768 2.504401 -0.3898946
## LLN         30.72376 2.520202 -0.3655381
## Virton      30.80012 2.124493 -0.2120129
```

Checking the goodness-of-fit:

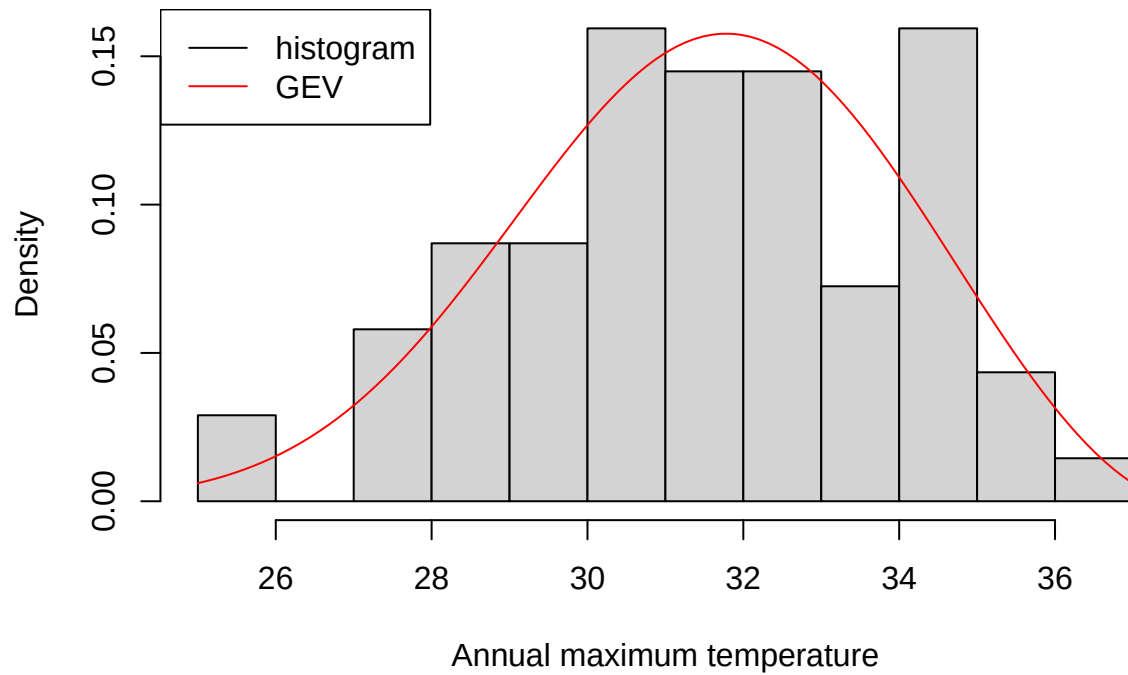
Annual maximum temperatures at Middelkerke (1950–2014)



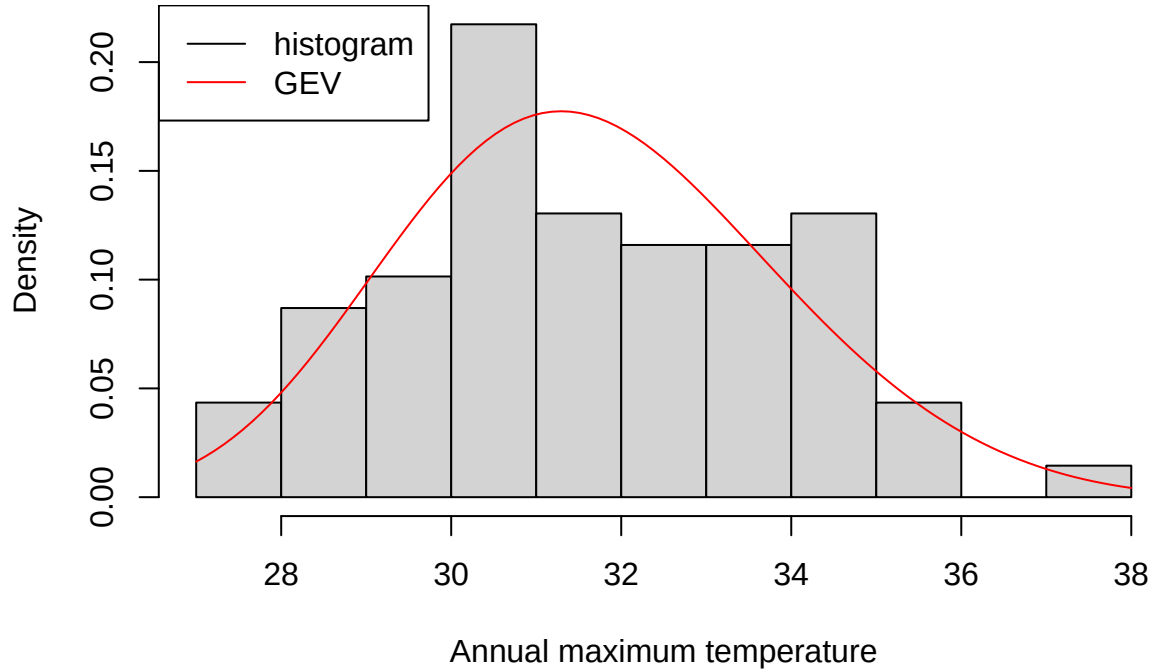
Annual maximum temperatures at Leuven (1950–2014)



Annual maximum temperatures at LLN (1950–2014)



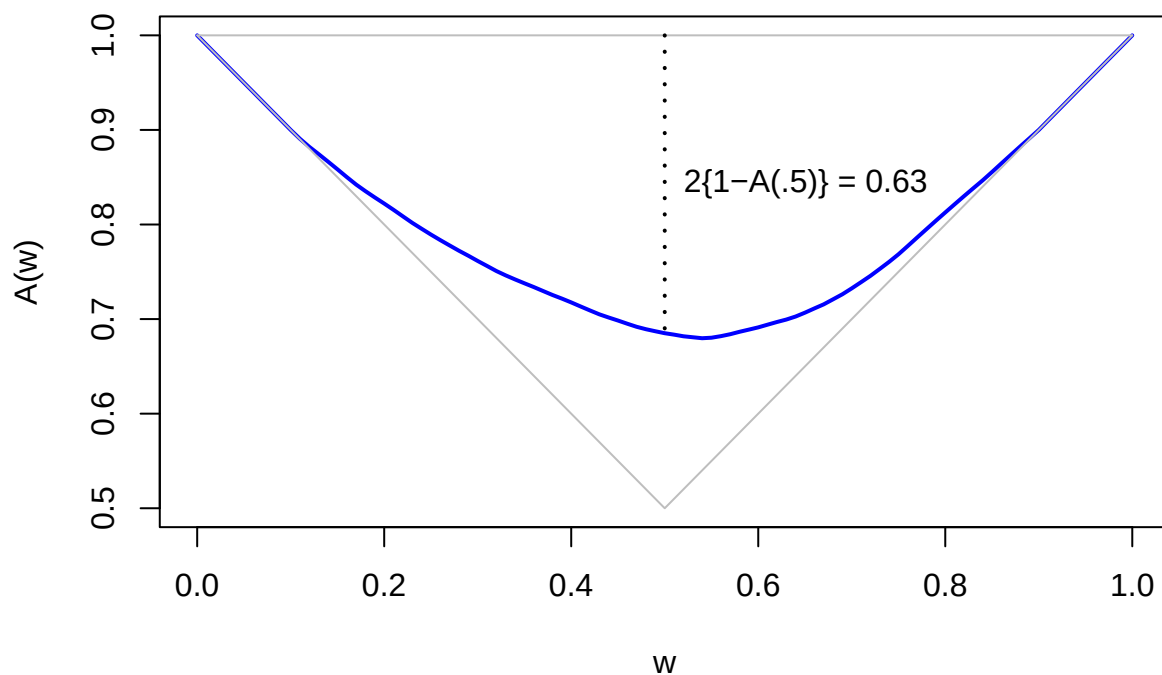
Annual maximum temperatures at Virton (1950–2014)



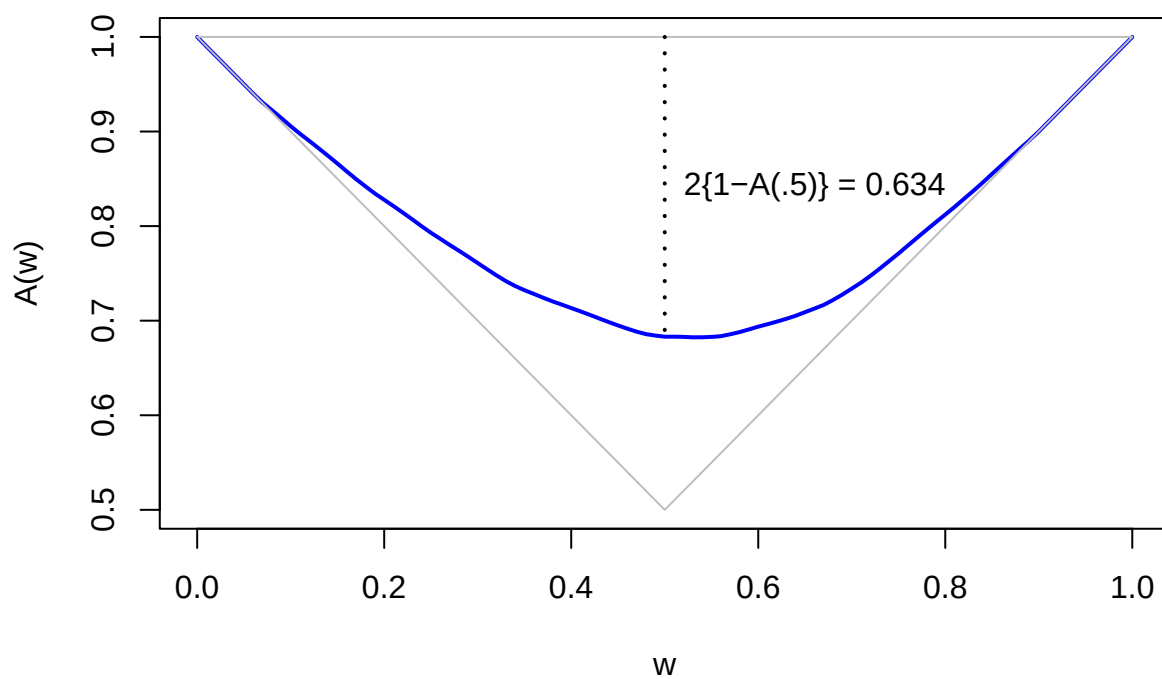
Pickands dependence functions

Nonparametric estimates of the bivariate Pickands dependence functions for each pair of cities:

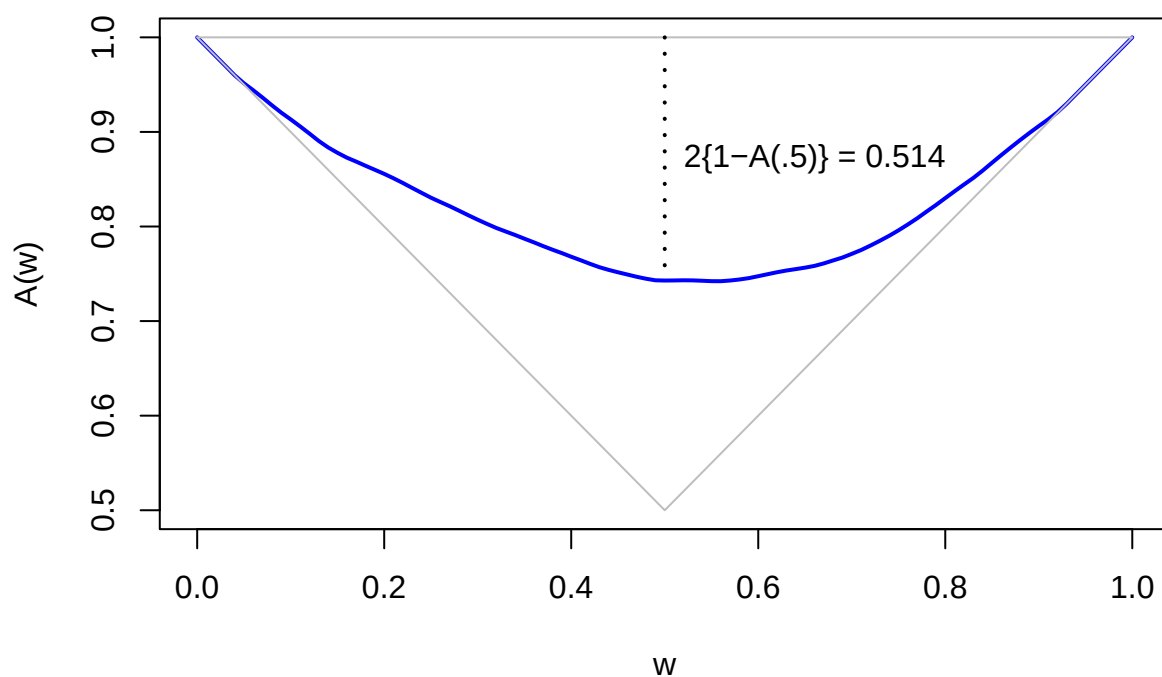
Pickands dependence function A, Middelkerke vs Leuven



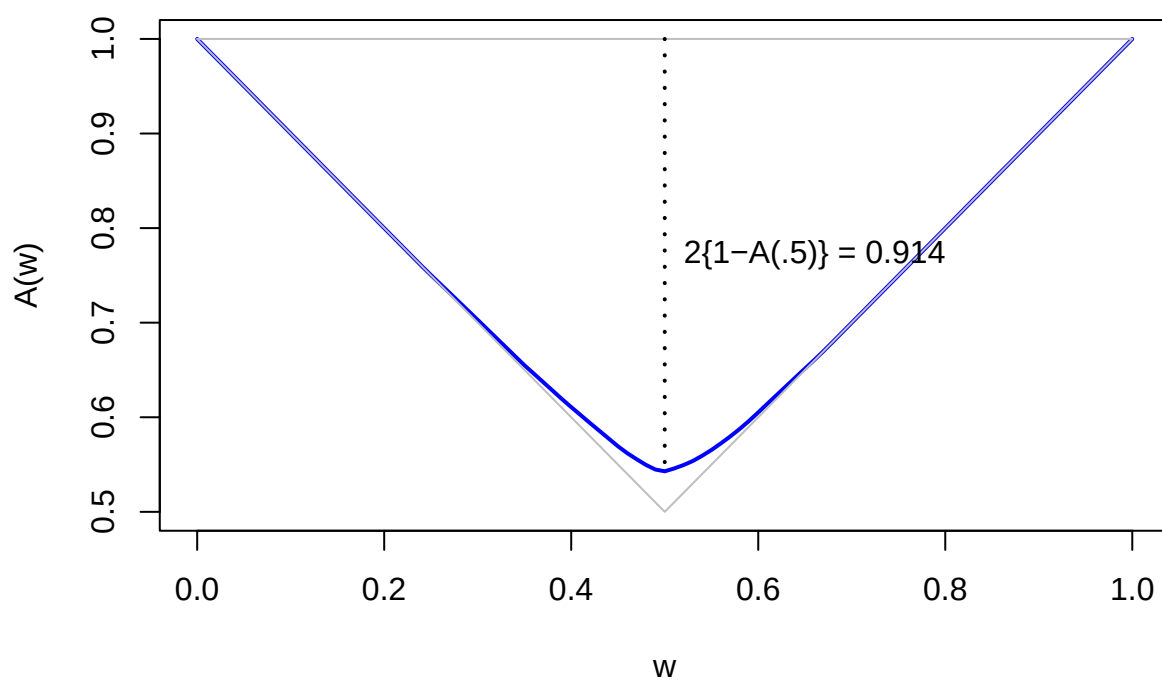
Pickands dependence function A, Middelkerke vs LLN



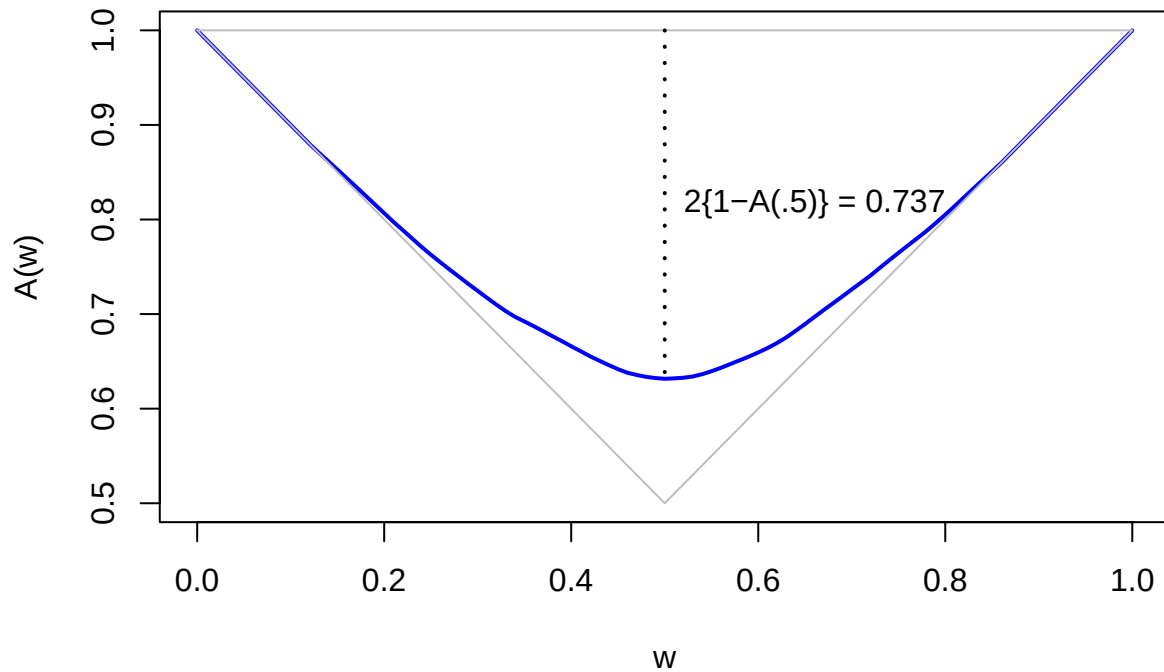
Pickands dependence function A, Middelkerke vs Virton



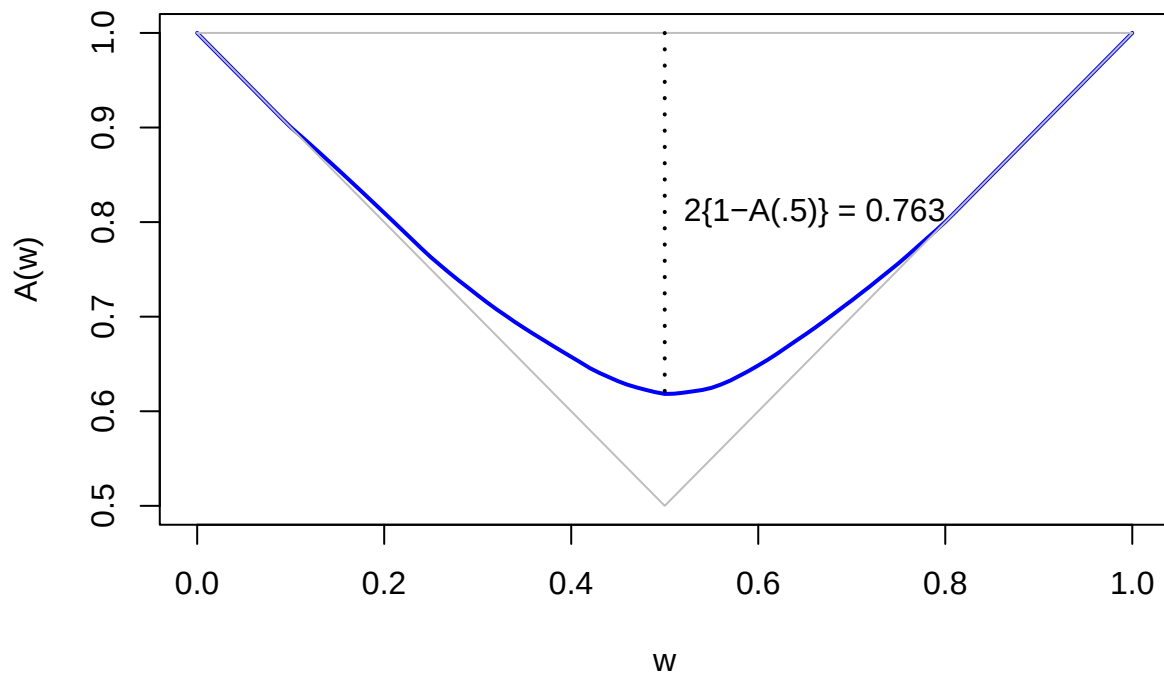
Pickands dependence function A, Leuven vs LLN



Pickands dependence function A, Leuven vs Virton



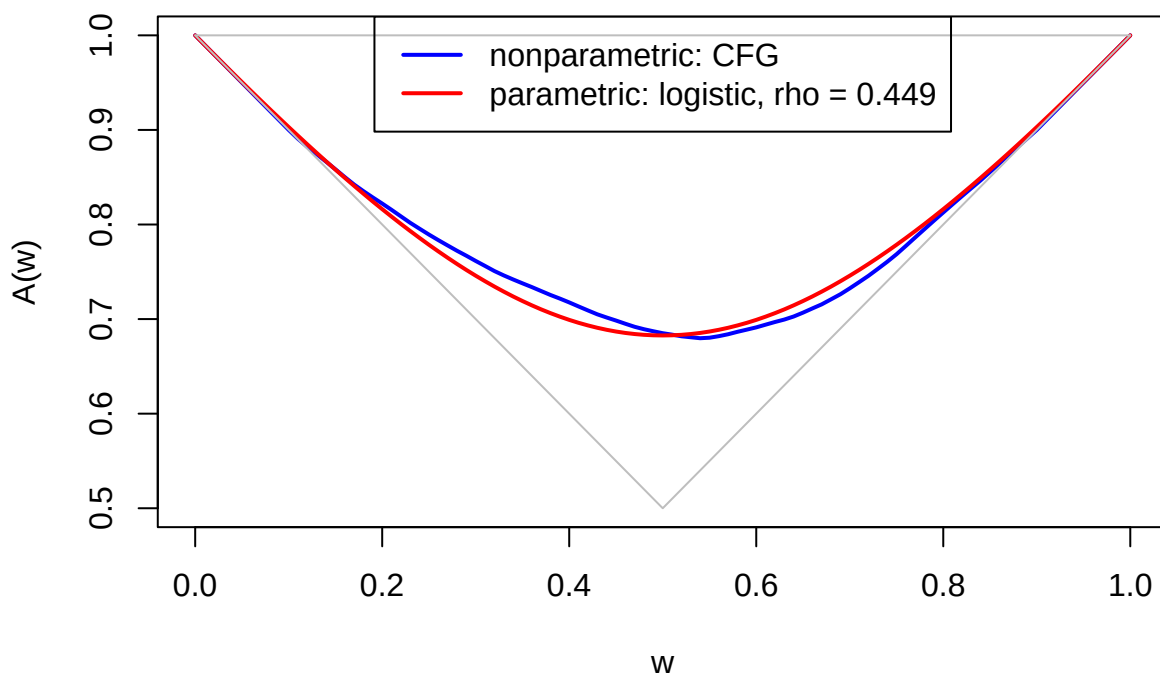
Pickands dependence function A, LLN vs Virton



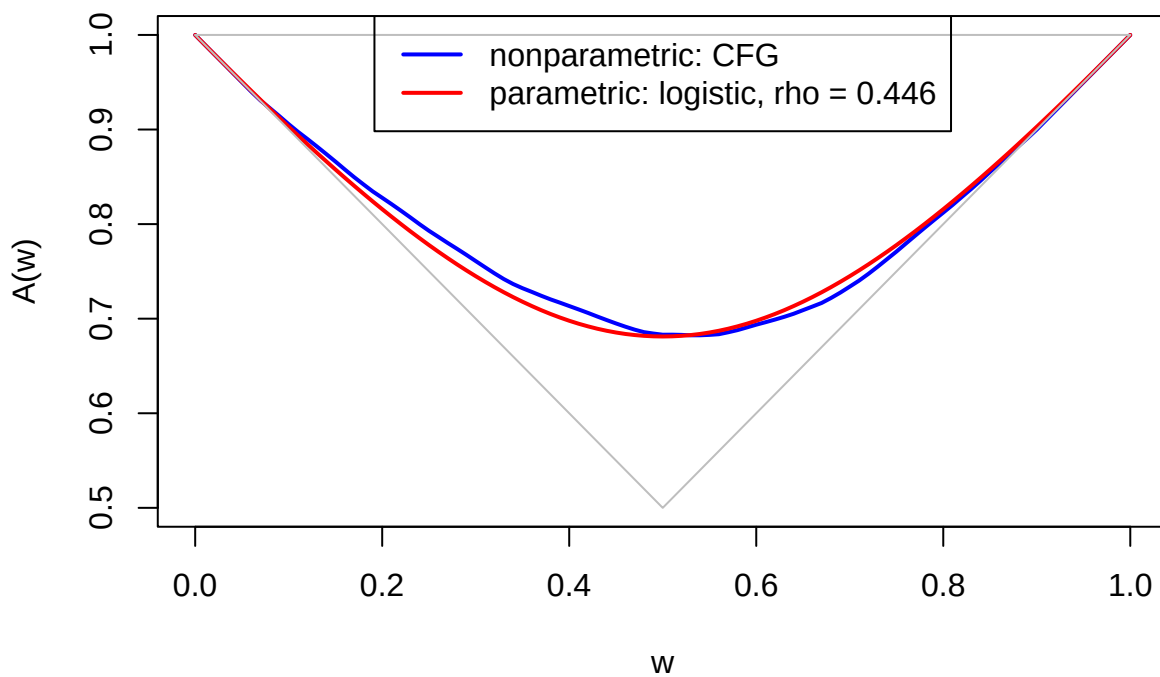
Maximum pseudo-likelihood

Fitting a Gumbel copula (= logistic model) by maximum pseudo-likelihood, using ranks, and comparing the obtained Pickands dependence functions with the nonparametric estimates:

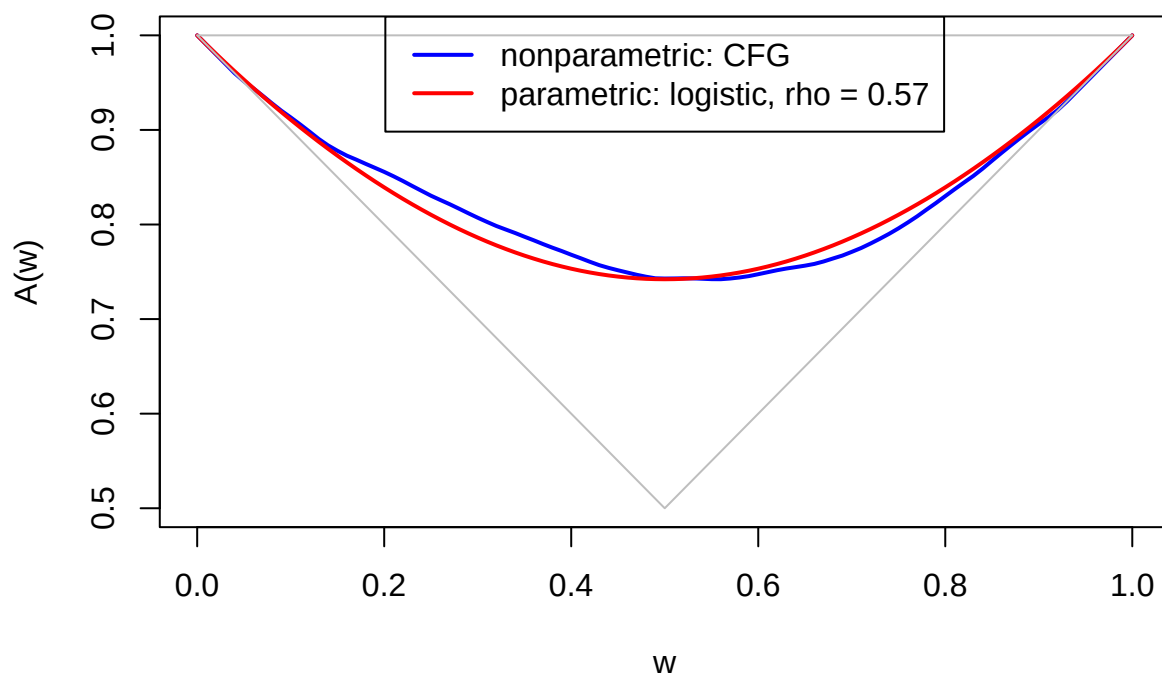
Pickands dependence function A, Middelkerke vs Leuven



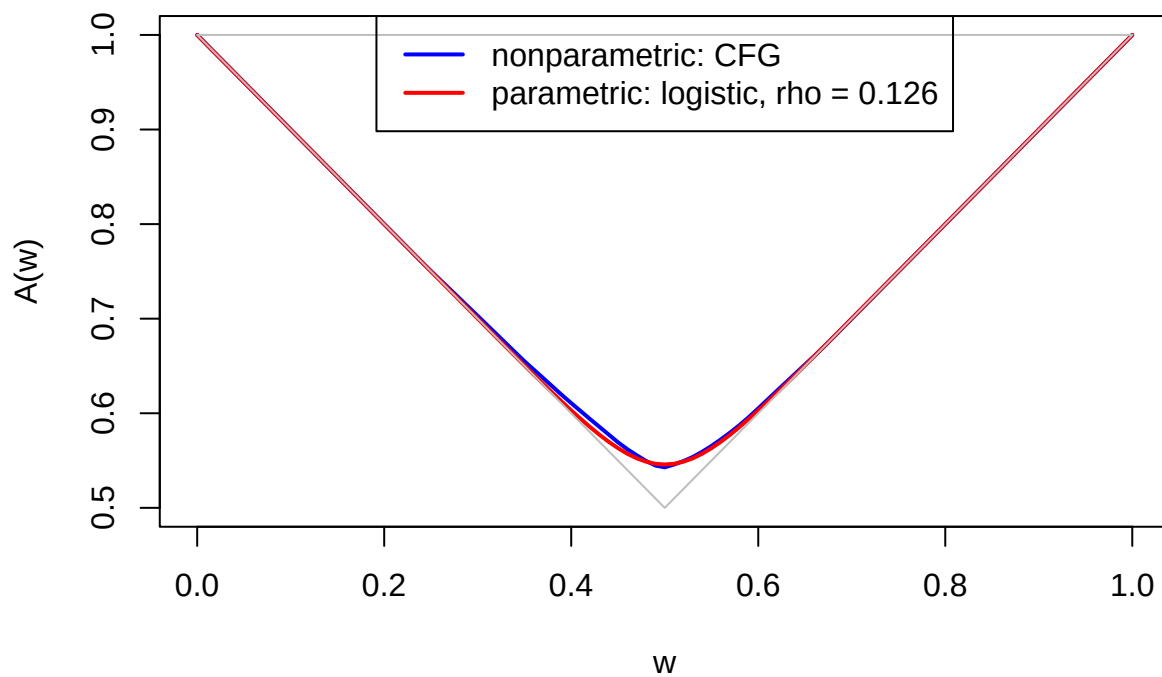
Pickands dependence function A, Middelkerke vs LLN



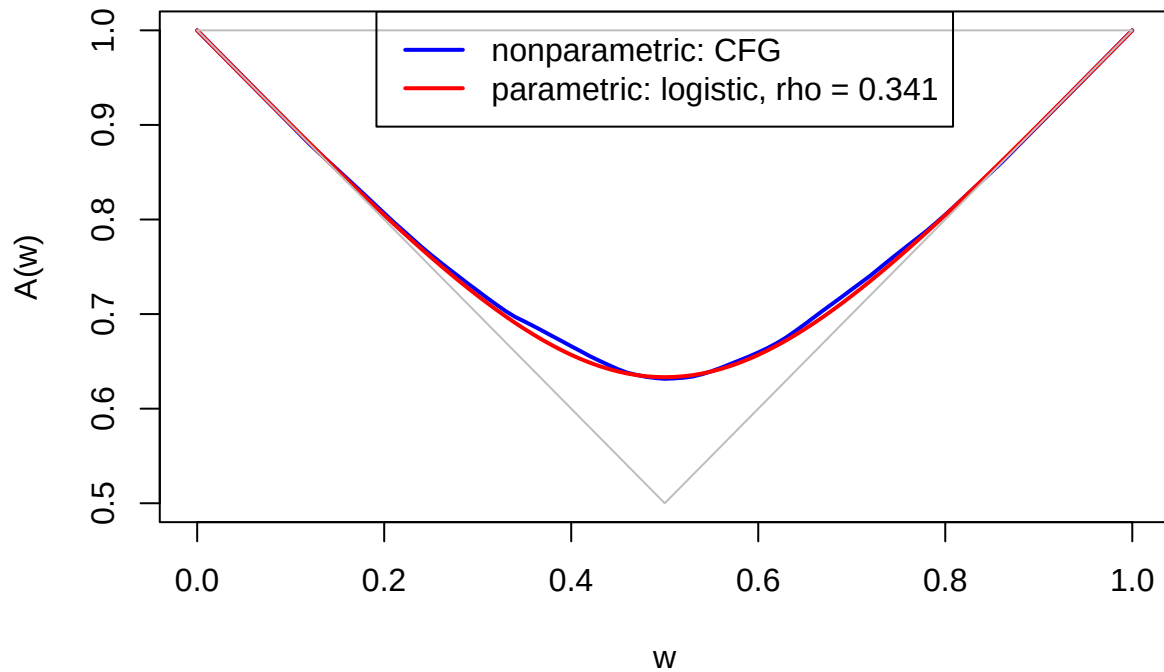
Pickands dependence function A, Middelkerke vs Virton



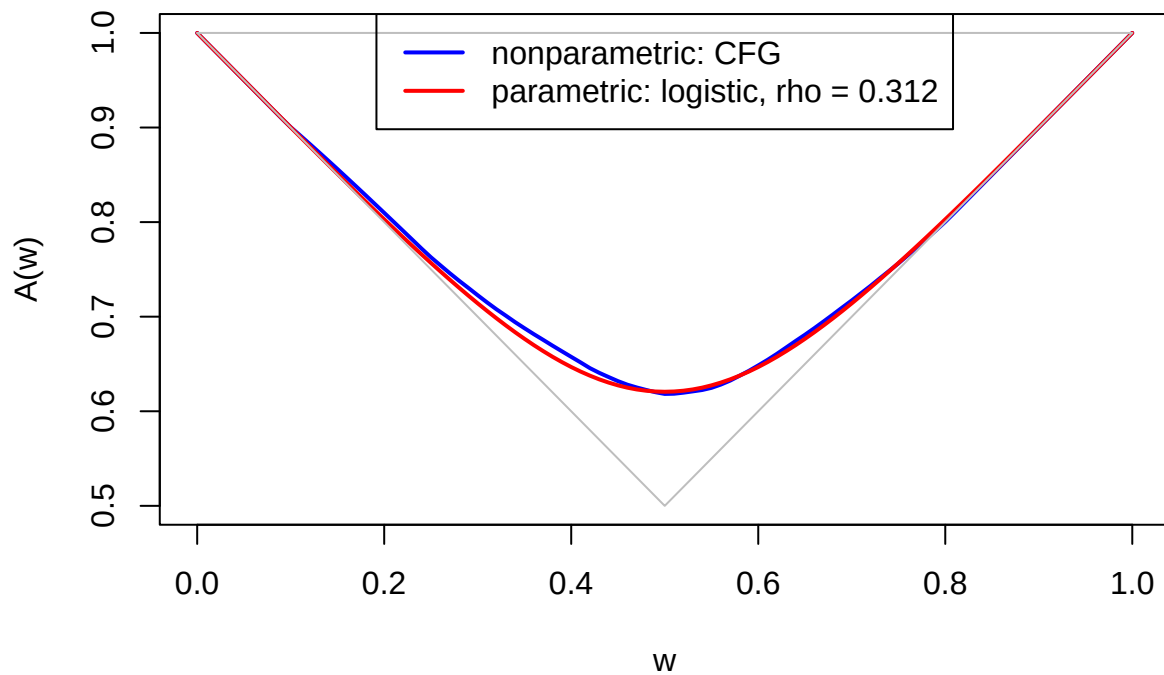
Pickands dependence function A, Leuven vs LLN



Pickands dependence function A, Leuven vs Virton



Pickands dependence function A, LLN vs Virton



Parameter estimation

Fitting a parametric bivariate GEV (margins + dependence) to a pair of cities:

##

```

## Attaching package: 'evd'

## The following objects are masked from 'package:mev':
##
##      dgev, pgev, qgev, rgev, venice
##
## Call: fbvevd(x = Temp[, c(30, 54)], model = "log")
## Deviance: 513.3078
## AIC: 527.3078
## Dependence: 0.7865444
##
## Estimates
##      loc1      scale1      shape1      loc2      scale2      shape2
## 30.5431685  2.4170041 -0.2034640  30.5623961  2.0306834 -0.0009132
##      dep
## 0.2791214
##
## Standard Errors
##      loc1  scale1  shape1      loc2  scale2  shape2      dep
## 0.31998  0.20729  0.07071  0.27442  0.18795  0.08236  0.03847
##
## Optimization Information
##   Convergence: successful
##   Function Evaluations: 46
##   Gradient Evaluations: 15
##
## Call: fbvevd(x = Temp[, c(30, 54)], model = "hr")
## Deviance: 512.7669
## AIC: 526.7669
## Dependence: 0.7814379
##
## Estimates
##      loc1      scale1      shape1      loc2      scale2      shape2      dep
## 30.500771  2.416119 -0.184874  30.586896  2.024898 -0.009385  3.604309
##
## Standard Errors
##      loc1  scale1  shape1      loc2  scale2  shape2      dep
## 0.31941  0.20763  0.06304  0.27193  0.18630  0.07719  0.52575
##
## Optimization Information
##   Convergence: successful
##   Function Evaluations: 41
##   Gradient Evaluations: 16

```